Question#1:
(CLO 6. Linked Lists: ________/14)

1.1 What is the functionality of the following piece of code for a given Singly-Linked List with first node as head **– See Fig. 1** **(2 points)**

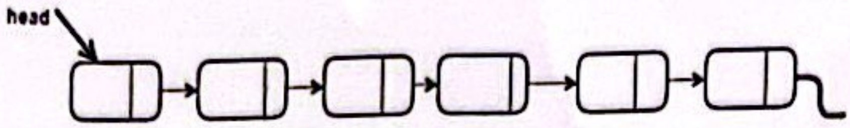


Fig.1

```
LLnode helpPtr= head;
int var= 0;
while (helpPtr!= null)            {
        if(helpPtr.data== key)
                return var;
        var= var+1;
        helpPtr= helpPtr.next;     }
```

a) Finds and deletes a given element in the list.
b) Finds and returns the given element in the list.
c) Finds and returns the position of the given element in the list.
d) Finds and inserts a new element in the list.

1.2 What are the contents of a given Singly-Linked List with first node as head **and contains the following data items: 2 → 3 → 4 → 5 → 6 after executing the function** HaveFun(). **(2 points)**

```
void HaveFun (){
LLnode ptr= head;
while (ptr.next.next!=  null ) {
      ptr= ptr.next;             }
ptr.next= null;  }
```

a) 2 → 3 → 4

b) 2 → 3 → 4 → 5

c) 2 → 3 → 4 → 5 → 6

d) null

1.3 Consider the following sorted Doubly-Linked List, where `head` is pointing to the first node in the list – See Fig. 2.

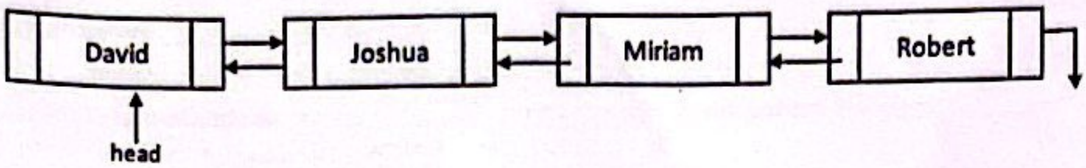


Fig. 2

What output is displayed as the result of executing the following statement? ________

```
system.out.println(head.next.prev.data);
```

(2 points)

1.4 Determine where the new node `NewNode` will be located after executing the following code segment? ________ **(2 points)**

```
DllNode NewNode= new  DllNode("Hanna");
DllNode Ptr= head;
while (Ptr.next!= null)
      Ptr=Ptr.next;
NewNode.prev = Ptr;
Ptr.next = NewNode;
```

1.5 Complete the following `exam01(ll_node head01, ll_node head02)` method that operates on an existing Singly-Linked List of positive and negative integers accessible by `head01`. Your method should copy only positive integer values to another empty Singly-Linked List accessible by `head02` (`head02` is `null` initially) – See Fig. 3 **(6 points)**

```
public void exam01(ll_node head01, ll_node head02) {
      ll_node hp = head01;
      ______________________
            if (hp.data > 0)
                  ______________________
            hp = hp . next;
      }
}
```

Question#2:
(CLO 7. Recursion: ________ /12)

2.1 For the given method call `count(10,20)`, **match the correct output (on the right) to the given** `void count(int a, int b)` **method (on the left).**

(4 points)

<pre>public static void count(int a, int b) {
if (a <= b)
       { count(a+1, b);
         System.out.print(a + " ");   }
}</pre>
	____	**a)**
		10 11 12 13 14 15 16 17 18 19 20

<pre>public static void count(int a, int b) {
if (a <= b)
       { System.out.print(a + " ");
         count(a+1, b);
         System.out.print(a + " ");   }
}</pre>
	____	**b)**
		10 11 12 13 14 15 16 17 18 19

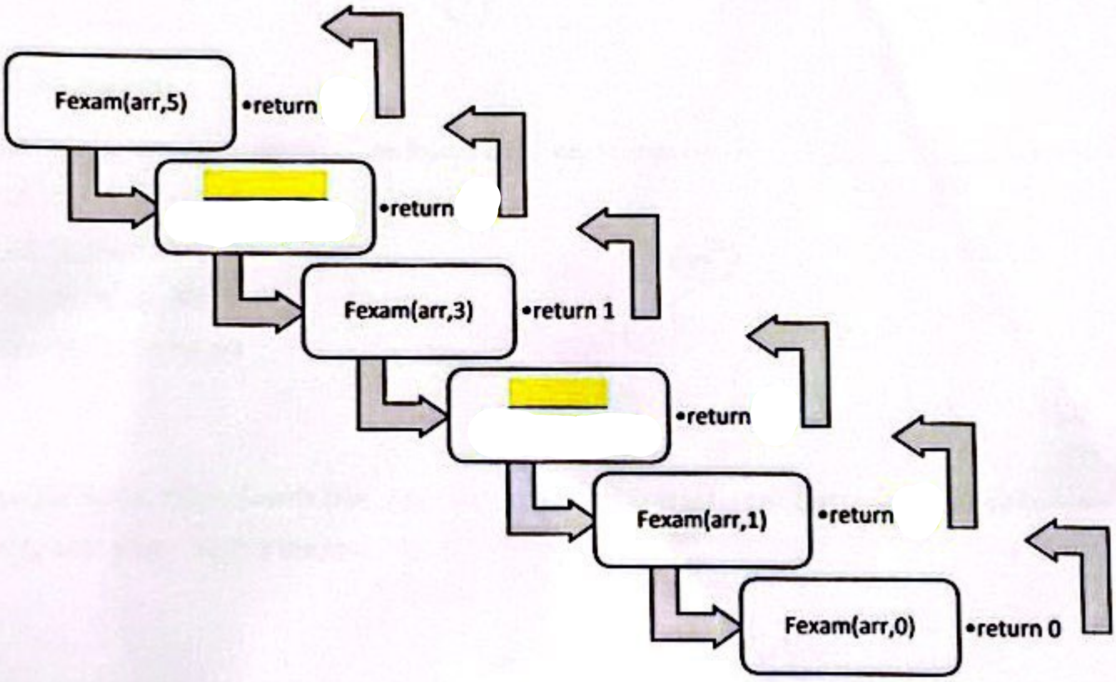
		c) **20 19 18 17 16 15 14 13 12 11 10**
		d) **10 11 12 13 14 15 16 17 18 19 20 20 19 18 17 16 15 14 13 12 11 10**
		e) **20 19 18 17 16 15 14 13 12 11 10 10 11 12 13 14 15 16 17 18 19 20**

2.2. Consider the following recursive function `Fexam(int [] arr, int size)`. Complete the following tracing for all recursive calls (the boxes) and show the final answer that is returned by the call `Fexam(num, 5)`.

`num` is an array of integer and containing the following values: `int num[] = {6, 9, 15, 1, 5}`

(8 points)

```
public static int Fexam(int [] arr, int size) {
    if (size== 0)
        return 0;
    else if (arr[size -1] % 5== 0)
        return 1 + Fexam(arr, size-1)
    else
        return Fexam(arr, size-1)
}
```



Each return → 1 → total 4

Each call → 2 → total 4

Question#3:
(CLO 11. Binary Search Trees: ________/12)

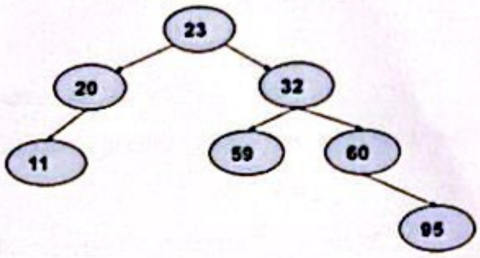
3.1 Construct a Binary Search Tree by inserting the following sequence of numbers: 10, 12, 5, 4, 20, 8, 7

(6 points)

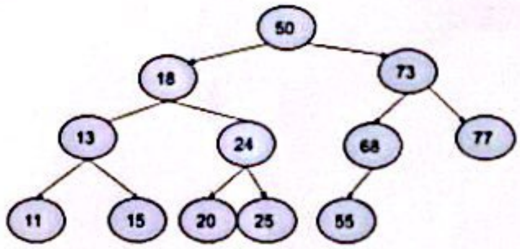
Each 1 except the root

3.2 The correct post-order traversal of the following Binary Search Tree is ______________ (2 points)

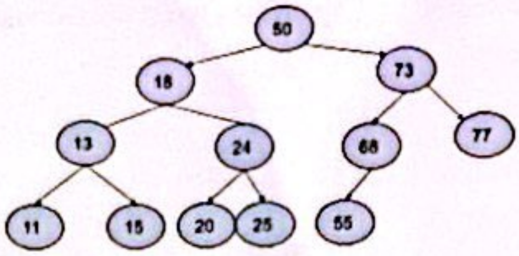
a) 23, 20, 11, 32, 59, 60, 95
b) 11, 20, 23, 59, 32, 60, 95
c) 11, 20, 59, 95, 60, 32, 23
d) 11, 20, 59, 32, 23, 60, 95



3.3 Given the below Binary Search Tree, draw only one of the two potential binary search trees that you can end up with if you remove the root (50). (2 points)



3.4 What is the content for below Binary Search Tree after executing the method `ExamTree(BSTNode p)` **assumeing that p is pointing to the** `root` **of the tree. Also what is the final return value.** **(2 points)**




```
public int ExamTree(BSTNode p) {
      if (p != null)   {
            if (p.left== null && p.right== null)
                    return p.data += 2;
            else
                    return p.data + ExamTree(p.right);
                    }
      }
```

a) No change of the nodes' values; it returns the sum of values of all nodes.

b) All leaf nodes are updated by adding 2 to their values; it returns the sum of values of all nodes of the right sub tree.

c) Only the most right node is updated by adding 2 to its value; it returns the sum of values of all nodes.

d) Only the most right node is updated by adding 2 to its value; it returns the sum of values of all nodes of the right sub tree.

Question#4:
(CLO 13. Algorithm Analysis: ________ /10)

4.1 What is the efficiency (running time) of the following algorithms in Big O( ) notation? **(6 points)**

a)

```
int Mystery1(int n){
int s = 0;
for (int i=1; i<= n; i++)
        for (int j=1; j <= 3; j++)
                s= s+i*i;
return s;
}
```

O(________)

b)

```
void Mystery2(){
for (int i=1; i<= n; i++)
        for (int j=n; j>= 1; j--)
                if (i>=n) system.out.println(i);
}
```

O(________)

c)

```
void Mystery3(){
for (int i=1; i<= n; i++){
        system.out.println("Exam");
        break;
}
}
```

O(________)

4.2 Give the Big O ( ) running times of the following operations in terms of n. **(4 points)**

Operation	O(n)	Operation	On)
Finding a value in a **Hash Table** whose hash function evenly distributes all items		Given an **Adjacency Matrix**, determine if an edge is between two specific nodes	
Finding the maximum value in a sorted array of *n* items		Sorting *n* elements using **Insertion Sort** (worst case)	

Question#5:
(CLO 14. Sorting Algorithms: ________/12)

5.1 Which sorting algorithm is described by: split a list into individual lists, then combine these, two lists at a time. (2 points)

a) Bubble
b) Merge
c) Insertion
d) None of these

5.2 Given the following list of numbers [54, 26, 93, 17, 77, 31, 44, 55, 20]. Show the contents of the list after the first partitioning according to the quick sort algorithm (choose first element as pivot)? Also shows the two halves (right and left lists) which will be invoked recursively by the quick sort. (9 points)

Left half

Right half

Each 2 numbers on the original list → 1 except the pivot 1 → total 5

Left and right lists → 4 (each 2)

5.3 Imagine that the following list of 10 numbers is to be sorted by quick sort: 3, 9, 5, 6, 0, 2, 7, 8, 4, 1
Which number would be the best pivot for the first iteration? (1 points)

a) 3
b) 9
c) 5
d) 0
e) 1

Question#6:
(CLO 8 and 9. Stack and Queue: ________ /20)

6.1 Let Q be a queue initially having the following integer values: 20, 40, 60, 80, 100, with 20 in the `front`. Also let S be a stack initially having the following integer values 11, 12 with 12 in the `top`.
What is the output of the following code segment? Also show the content of the stack S and queue Q.
(11 points)

```
public void funQueue () {
 //S is a Stack where top points at 12
 int[] S = {11, 12};
 //Q is a Queue where front points at 20
 int[] Q = {20, 40, 60, 80, 100};

for (int i = 0; i < 2; i++) {
      system.out.print(Q.dequeue() + " ");
      system.out.print(S.pop() + " ");
      Q.enqueue(S.top()+ i);
      S.push(Q.front());
   }
}
```

Output: ____________________

Content of S: ____________________

Content of Q: ____________________

Each 1

Output → 4 (each 1)

Content of S → 2 (each 1)

Content of Q → 5 (each 1)

6.2 Suppose we have a circular array implementation of the queue, with ten items in the queue stored at data[2] through data[11], the current capacity is 12. Where does the insert method place the new entry in the array? **(1 points)**

a) data[1]
b) data[0]
c) data[11]
d) data[12]

6.3 Evaluate the following postfix expression using a Stack. Additionally, you must show the contents of the Stack at the indicated points (A, B, and C) in the given postfix expression. Do not forget to write the final result.

(8 points)

A B C

2 13 + 5 - 6 3 / 5 * +

A (After first +), B (After /), and C (After *)

A

B

C

Expression Value (Final Result):

Each value, all Stack points → 5 (each 1)

Final result → 3

Question#7:
(CLO 15. Hash Table: ________/10)

7.1 Given the following keys (12, 18, 13, 3, 23), and the length of the hash table is 10.
What is the result of the hash table if the hash function h(x) = x mod 10 is used with Quadratic Probing and Separate Chaining. **(10 points)**

Quadratic Probing	
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	

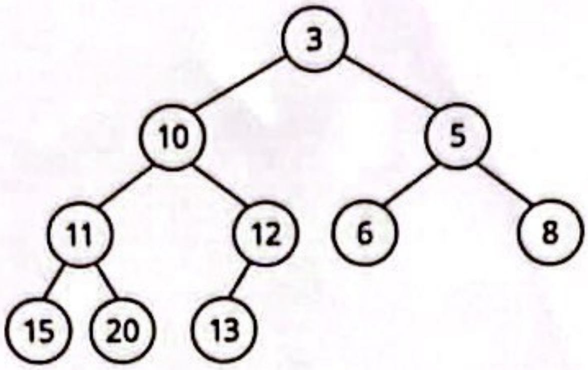
5 (each 1)

Separate Chaining	
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	

5 (each 1)

Question#8:
(CLO 16. Heap: __________ /10)

8.1 Consider the following min Heap, Insert the value 7 then show the final Heap tree content after inserting. **(10 points)**



Extra leaf in the right place→ 2

Exchange (12, 7) → 4 (7,10) → 4